

What's so Universal about Universal Responses?

On Violations of Transfer-Kernel Factorisation in Unfolding Problems

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TL;DR

Likelihood-based and reweighting-based unfolding methods in high energy physics assume that the detector response is *universal*, that is, that the conditional probability density of the detector-level event given the particle-level event is the same in nature and in simulation. This note demonstrates that this assumption is representation dependent.

Universality of the response conditional on the complete particle-level state neither implies nor is implied by universality of the response induced on a chosen pair of particle- and detector-level summaries, and neither property implies identifiability of a particle-level observable from detector-level data. This note characterises the induced response exactly as a mixture of the full response over the generator-dependent conditional law of the omitted particle-level information, and gives necessary and sufficient conditions for universality to survive coarse-graining. It then shows that induced universality is equivalent to a conditional independence statement whose violation is measured by a conditional mutual information, and that this quantity equals the population gain in binary cross-entropy between two nested Bayes optimal classifiers.

Applying the resulting test to paired HERWIG+DELPHES and PYTHIA+DELPHES samples [1], one finds that the detector-level value carries generator information beyond the corresponding particle-level value for ungroomed jet mass and, with decreasing confidence, for four of the other five jet substructure observables unfolded in Andreassen et al. [2]; the evidence for jet width is inconclusive. The effect is largest for ungroomed jet mass, (9.061 ± 0.465) mnat/event, which is 4.2 times the next-largest observable. This finding is mathematically compatible with, but not

diagnostic of, the particle–identity explanation offered in Andreassen et al. [2] for the poor unfolding performance on jet mass. The note proposes conditional tests that separate response misspecification from non-identifiability.

1 Introduction.

Unfolding, the estimation of a particle–level distribution from detector–level observations, requires a model of the detector response [3, 4]. In practice, the response is learned from simulated pairs of particle– and detector–level events and is then applied to measured data. This transfer is valid only if the conditional distribution of the detector–level event given the particle–level event is the same in simulation as in nature. This is referred to as *response universality*.

It is assumed, explicitly or implicitly, by iterative Bayesian unfolding [5], by regularized matrix inversion methods, and by machine learning methods such as OMNIFOLD [2, 6].

The purpose of this note is to separate three properties that may be conflated:

- (U1) *universality of the full response*, i.e., of the detector response conditional on the complete particle–level state;
- (U2) *universality of an induced response*, i.e., of the response obtained after particle– and detector–level events are projected to chosen summary observables;
- (U3) *identifiability* of the distribution of a particle–level observable from the distribution of detector–level events.

This note will show that property (U1) does not imply property (U2), that property (U2) does not imply property (U1), and that property (U1) does not imply property (U3).

1.1 Summary of results.

1. The induced response on a pair of summaries is the full response averaged against the generator dependent conditional law of the particle–level information discarded by the summary (lemma 2). Consequently, the induced response fails to be universal precisely when the generators disagree on this conditional law in directions that the projected detector response does not annihilate (theorem 1). An explicit model in which the full response is universal but the

induced response is not universal at every value of the summary is given in example 1.

2. Induced universality is equivalent to the conditional independence $X \perp\!\!\!\perp S \mid Z$ between the detector-level summary X and a generator label S , given the particle-level summary Z (proposition 3). Its failure is quantified by the conditional mutual information $I(S; X \mid Z)$, which is a weighted Jensen–Shannon divergence between the two induced responses (proposition 4) and equals the difference in population binary cross-entropy between Bayes optimal classifiers of S from Z and from (Z, X) (proposition 5). For learned classifiers, the population gain differs from $I(S; X \mid Z)$ precisely by the difference of excess risks, which (corollary 2) makes explicit.
3. Applied to the benchmark samples of Andreassen et al. [1], all six scalar observables exhibit a positive held-out gain, although only for jet mass does the gain exceed the run-to-run variability of a constructed null by a wide margin. The gain for ungroomed jet mass is approximately four times the next-largest value and sixteen times the smallest (table 1). This constitutes strong evidence that the induced mass response is not universal between HERWIG and PYTHIA, notwithstanding the use of an identical detector simulation.
4. This finding does not establish $Y \not\perp\!\!\!\perp S \mid T$, where T and Y denote the full particle- and detector-level representations supplied to OMNIFOLD. The explanation given in Andreassen et al. [2] for the mass residual, namely that particle identity affects jet mass but is only partially observed at detector-level, is a mechanism that produces induced non-universality and non-identifiability simultaneously, even under an exactly universal full response (proposition 9).

However, the within-generator technical closure test reported in Andreassen et al. [2] has no power against a cross-generator violation of the full response (proposition 10). The attribution of the mass residual therefore remains empirically open. This note proposes tests that resolve it (section 8).

1.2 Outline.

Section 2 fixes notation and definitions. Section 3 analyses the effect of coarse-graining on universality. Section 4 develops the classification based test. Section 5 reports the empirical study. Section 6 derives the consequences for unfolding in a reduced space and for identifiability in the full space. Section 7 applies these results

to the OMNIFOLD jet mass result, and section 8 proposes follow-up tests. All proofs are collected in appendix A.

2 Setup and definitions.

Throughout, this note

$T \in \mathcal{T}$ denotes the complete particle-level representation of an event,

$Y \in \mathcal{Y}$ denotes the complete detector-level representation,

$Z = f(T) \in \mathcal{Z}$ denotes a particle-level summary, and

$X = g(Y) \in \mathcal{X}$ denotes the corresponding detector-level summary.

For example, Z and X may be the particle- and detector-level jet mass, respectively. The generator label takes values in $\mathcal{S} = \{\text{H}, \text{P}\}$, with HERWIG playing the role of nature and PYTHIA the role of simulation.

For any topological space $\mathcal{V}$, let $\mathcal{B}(\mathcal{V})$ denote its Borel σ -algebra (the collection of measurable subsets of $\mathcal{V}$). When $(\mathcal{V}, \mathcal{B}(\mathcal{V}))$ is a standard Borel space, let $\mathcal{P}(\mathcal{V})$ denote the convex set of Borel probability measures on $(\mathcal{V}, \mathcal{B}(\mathcal{V}))$, and let $\mathcal{M}(\mathcal{V})$ denote the Banach space of finite signed Borel measures on $(\mathcal{V}, \mathcal{B}(\mathcal{V}))$ equipped with the total variation norm.

Assumption 1 (Measurable structure). The sample spaces $\mathcal{T}, \mathcal{Y}, \mathcal{Z}, \mathcal{X}$ are standard Borel spaces (each equipped with its respective Borel σ -algebra $\mathcal{B}(\cdot)$), and the maps $f: \mathcal{T} \rightarrow \mathcal{Z}$ and $g: \mathcal{Y} \rightarrow \mathcal{X}$ are Borel measurable.

For $s \in \mathcal{S}$, let $\mathbb{P}_s \in \mathcal{P}(\mathcal{T} \times \mathcal{Y})$ be the joint law of (T, Y) under generator s , let $Q_s \in \mathcal{P}(\mathcal{T})$ be the marginal law of T , and let $\mu_s = f_{\#}Q_s \in \mathcal{P}(\mathcal{Z})$ be the induced law of $Z = f(T)$.

Lemma 1 (Regular conditional distributions and disintegration [7]). *Under assumption 1, $\forall s \in \mathcal{S} \exists$ Markov kernels*

$$K_s : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{Y}), \quad r_s : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X}), \quad \text{and} \quad Q_s^{(\cdot)} : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{T})$$

$$\forall B \in \mathcal{B}(\mathcal{T}) \quad \forall A \in \mathcal{B}(\mathcal{Y}) \quad \forall B' \in \mathcal{B}(\mathcal{Z}) \quad \forall C \in \mathcal{B}(\mathcal{X})$$

$$\mathbb{P}_s(T \in B, Y \in A) = \int_B K_s(A | t) Q_s(dt), \quad (1)$$

$$\mathbb{P}_s(Z \in B', X \in C) = \int_{B'} r_s(C | z) \mu_s(dz), \quad (2)$$

$$\mathbb{P}_s(Z \in B', T \in B) = \int_{B'} Q_s^z(B) \mu_s(dz). \quad (3)$$

Each kernel is uniquely determined up to a null set of its driving marginal (Q_s , μ_s , and μ_s , respectively). Furthermore,

$$\mu_s \left\{ z \in \mathcal{Z} \mid Q_s^z(f^{-1}(\{z\})) = 1 \right\} = 1. \quad (4)$$

Definition 1 (Full and induced response). $\forall s \in \mathcal{S}$, K_s of eq. (1) is the *full response* of generator s , and r_s of eq. (2) is the *response induced on* (f, g) . Q_s^z of eq. (3) is the *conditional composition law* of generator s at $Z = z$.

Since a conditional law is determined only up to null sets of the conditioning marginal, universality is properly defined as the existence of a common kernel.

Definition 2 (Universality).

1. The full response is *universal* across $\mathcal{S}$ if

$$\begin{aligned} \exists K : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{Y}) \quad \forall s \in \mathcal{S} \quad \forall B \in \mathcal{B}(\mathcal{T}) \quad \forall A \in \mathcal{B}(\mathcal{Y}) \\ \mathbb{P}_s(T \in B, Y \in A) = \int_B K(A | t) Q_s(dt). \end{aligned} \quad (5)$$

That is to say,

$$\forall s \in \mathcal{S} \quad Q_s \{ t \in \mathcal{T} \mid K(\cdot | t) = K_s(\cdot | t) \} = 1 \quad (6)$$

2. The induced response on (f, g) is *universal* across $\mathcal{S}$ if

$$\begin{aligned} \exists r : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X}) \quad \forall s \in \mathcal{S} \quad \forall B' \in \mathcal{B}(\mathcal{Z}) \quad \forall C \in \mathcal{B}(\mathcal{X}) \\ \mathbb{P}_s(Z \in B', X \in C) = \int_{B'} r(C | z) \mu_s(dz). \end{aligned} \quad (7)$$

That is to say,

$$\forall s \in \mathcal{S} \quad \mu_s \{ z \in \mathcal{Z} \mid r(\cdot | z) = r_s(\cdot | z) \} = 1 \quad (8)$$

3. The g -projected full response is *universal* across $\mathcal{S}$ if

$$\begin{aligned} \exists K^g : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{X}) \quad \forall s \in \mathcal{S} \quad \forall B \in \mathcal{B}(\mathcal{T}) \quad \forall C \in \mathcal{B}(\mathcal{X}) \\ \mathbb{P}_s(T \in B, X \in C) = \int_B K^g(C \mid t) Q_s(dt). \end{aligned} \quad (9)$$

If the full response is universal with common version K , then the g -projected full response is universal with common version

$$K^g(A \mid t) = K[g^{-1}(A) \mid t]. \quad (10)$$

The converse is false in general. The assumption made in the likelihood derivation of OMNI_{FOLD} is universality of the full response in the sense of definition 2 (see eq. (57)).

Remark 1 (Common essential support and non-uniqueness). $\forall s \in \mathcal{S} \quad \nu = \frac{\mu_H + \mu_P}{2}, m_s = \frac{d\mu_s}{d\nu}$ is Borel versions of the Radon–Nikodym derivatives, and

$$\mathcal{Z}_\cap = \{z \in \mathcal{Z} \mid m_H(z) > 0 \wedge m_P(z) > 0\} \quad (11)$$

denotes the common essential support of μ_H and μ_P (well-defined up to a set of measure 0 under ν). The induced response is universal across $\mathcal{S}$ if and only if

$$\nu\left\{z \in \mathcal{Z}_\cap \mid r_H(\cdot \mid z) = r_P(\cdot \mid z)\right\} = 1. \quad (12)$$

Indeed, any common version r must satisfy

$$\forall s \in \mathcal{S} \quad \mu_s\left\{z \in \mathcal{Z}_\cap \mid r(\cdot \mid z) = r_s(\cdot \mid z)\right\} = 1. \quad (13)$$

Since μ_H, μ_P, ν are mutually absolutely continuous on $\mathcal{Z}_\cap$, they share identical null sets on $\mathcal{Z}_\cap$, forcing agreement ν -a.e. on $\mathcal{Z}_\cap$. Conversely, if r_H and r_P agree ν -a.e. on $\mathcal{Z}_\cap$, the composite kernel defined by

$$r(\cdot \mid z) = \begin{cases} r_H(\cdot \mid z) & \text{if } m_H(z) > 0, \\ r_P(\cdot \mid z) & \text{if } m_H(z) = 0 \end{cases} \quad (14)$$

defines a valid Markov kernel that serves as a common version for both generators. Consequently, universality imposes no constraints on the behaviour of the response kernels outside $\mathcal{Z}_\cap$. An identical equivalence holds for the full response with respect to Q_H, Q_P , and their mixture on $\mathcal{T}$.

Definition 3 (Forward operator). For every $\lambda \in \mathcal{M}(\mathcal{V})$ and every Markov kernel $\kappa : \mathcal{V} \rightarrow \mathcal{W}$, κ acts on λ by

$$\forall A \in \mathcal{B}(\mathcal{W}) \quad (\kappa\lambda)(A) = \int_{\mathcal{V}} \kappa(A \mid v) \lambda(\mathrm{d}v). \quad (15)$$

The map $\lambda \mapsto \kappa\lambda$ is linear, maps $\mathcal{P}(\mathcal{V})$ into $\mathcal{P}(\mathcal{W})$, and is a contraction in total variation. Its null space is denoted $\ker \kappa \subseteq \mathcal{M}(\mathcal{V})$.

3 Universality under coarse-graining.

Lemma 2 (Mixture representation of the induced response). *If the g -projected full response is universal across $\mathcal{S}$ with common version $K^g : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{X})$ then* Proof on [page 30](#).

$$\forall s \in \mathcal{S} \quad z \mapsto \int_{\mathcal{T}} K^g(\cdot \mid t) Q_s^z(\mathrm{d}t) \text{ is a version of } r_s : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X}), \quad (16)$$

where r_s is the induced response kernel and Q_s^z is the disintegration of Q_s along the fibres of f . In particular, if $\mathcal{T} = \mathcal{Z} \times \mathcal{U}$ where $\mathcal{U}$ is a standard Borel space collecting the latent particle-level coordinates omitted by Z , so that $f(z, u) = z$, and if $p_s(\cdot \mid z) \in \mathcal{P}(\mathcal{U})$ denotes the conditional distribution of $U \mid Z = z$ under Q_s , then

$$\forall s \in \mathcal{S} \quad \forall C \in \mathcal{B}(\mathcal{X}) \quad \mu_s \left\{ z \in \mathcal{Z} \mid r_s(C \mid z) = \int_{\mathcal{U}} K^g(C \mid z, u) p_s(\mathrm{d}u \mid z) \right\} = 1 \quad (17)$$

Lemma 2 shows that the induced response is not an intrinsic property of the detector alone. Rather, it is a functional of the pair $(K^g, Q_s^{(\cdot)})$, in which the second argument depends explicitly on the event generator.

Theorem 1 (Universality under coarse-graining). *If the g -projected full response is universal with common version K^g then* Proof on [page 30](#).

$$\nu \left\{ z \in \mathcal{Z}_{\cap} \mid r_H(\cdot \mid z) - r_P(\cdot \mid z) = K^g(Q_H^z - Q_P^z) \right\} = 1, \quad (18)$$

where the linear operator $K^g : \mathcal{M}(\mathcal{T}) \rightarrow \mathcal{M}(\mathcal{X})$ acts by

$$(K^g\eta)(C) = \int_{\mathcal{T}} K^g(C \mid t) \eta(\mathrm{d}t). \quad (19)$$

Consequently:

1 (a) r is universal across $\mathcal{S} \iff \nu\{z \in \mathcal{Z}_\cap \mid Q_H^z - Q_P^z \in \ker K^g\} = 1$

1 (b) Agreement of the generator conditional laws along the fibres of f is sufficient:

$$\nu\{z \in \mathcal{Z}_\cap \mid Q_H^z = Q_P^z\} = 1 \implies r \text{ is universal across } \mathcal{S}. \quad (20)$$

1 (c)

$$\begin{aligned} & \nu\left\{z \in \mathcal{Z}_\cap \mid K^g : \mathcal{M}(f^{-1}\{z\}) \hookrightarrow \mathcal{M}(\mathcal{X})\right\} = 1 \\ & \implies \left[r \text{ is universal across } \mathcal{S} \iff \nu\{z \in \mathcal{Z}_\cap \mid Q_H^z = Q_P^z\} = 1.\right] \end{aligned} \quad (21)$$

Theorem 1 (a) makes precise the two ways in which universality can survive coarse-graining when the generators disagree on composition. Either the projected detector response is insensitive to every direction in which the conditional composition laws differ, or its action on the difference cancels. The following sufficient condition corresponds to the first case holding uniformly.

Proposition 1 (Response-sufficient summaries). *If the g -projected full response is universal with common version K^g , and if* Proof on page 31.

$$\exists \text{ Markov kernel } \kappa : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X}) \quad \forall s \in \mathcal{S} \quad Q_s\left\{t \mid K^g(\cdot \mid t) = \kappa(\cdot \mid f(t))\right\} = 1, \quad (22)$$

then κ is a common version of r_H and r_P .

In particular, the induced response is universal for every pair of generator laws (Q_H, Q_P) satisfying eq. (22).

Definition 4 (Response sufficient summary). A summary f is defined to be *response-sufficient* for g if it satisfies eq. (22).

Jet observables are, in general, not response-sufficient. The following example, shows that the failure of induced universality can be total even when the full response is exactly universal.

Example 1 (Latent-species model). Fix a shift parameter $a \neq 0$, detector resolution $\sigma > 0$, and distinct generator fractions $\pi_H \neq \pi_P$ in $(0, 1)$.

Particle-level Let $\mathcal{T} = \mathbb{R} \times \{0, 1\}$ and $T = (V, U)$, where under generator $s \in \mathcal{S}$, the kinematic feature $V \sim \mathcal{N}(0, 1)$ and the latent particle species $U \sim \text{Bern}(\pi_s)$ are independent. The particle-level summary observable is

$$Z = f(V, U) = V + aU. \quad (23)$$

Detector-level Let $\mathcal{Y} = \mathcal{X} = \mathbb{R}$, let $g: \mathcal{Y} \rightarrow \mathcal{X}$ be the identity map, and suppose the detector measures only the raw kinematics V with Gaussian smearing:

$$K(\cdot \mid v, u) = \mathcal{N}(v, \sigma^2). \quad (24)$$

The full response, K , is universal across $\mathcal{S}$ by construction and is completely blind to the species label U .

However, the induced response on Z is *not* universal. By lemma 2, along the fibre $f^{-1}(\{z\})$, $V = z - aU$, which gives the mixture

$$r_s(\cdot \mid z) = \sum_{u \in \{0,1\}} p_s(u \mid z) \mathcal{N}(z - au, \sigma^2), \quad (25)$$

where the posterior species probability under generator s is

$$w_s(z) = p_s(1 \mid z) = \mathbb{P}_s(U = 1 \mid Z = z) = \frac{\pi_s \phi(z - a)}{\pi_s \phi(z - a) + (1 - \pi_s) \phi(z)}, \quad (26)$$

with ϕ denoting the standard Gaussian density. Because $\pi_H \neq \pi_P$ and $a \neq 0$, we have $p_H(1 \mid z) \neq p_P(1 \mid z)$ for almost every z . Consequently,

$$\forall z \in \mathcal{Z} \quad r_H(\cdot \mid z) \neq r_P(\cdot \mid z), \quad (27)$$

demonstrating that universality of the full detector response does not imply universality of the induced observable response.

Lemma 3 (Properties of the latent-species model). *For the construction in example 1* Proof on page 31.

3 (a) *The full response is universal.*

3 (b) *μ_H and μ_P are equivalent to Lebesgue measure, so $\mathcal{Z}_\cap = \mathbb{R}$.*

3 (c)

$$\begin{aligned} \forall z \in \mathbb{R} \quad r_s(\cdot \mid z) &= [1 - w_s(z)] \mathcal{N}(z, \sigma^2) + w_s(z) \mathcal{N}(z - a, \sigma^2) \\ &\wedge r_H(\cdot \mid z) \neq r_P(\cdot \mid z). \end{aligned} \quad (28)$$

3 (d) *$KQ_H = KQ_P = \mathcal{N}(0, 1 + \sigma^2)$, whereas $f_\#Q_H \neq f_\#Q_P$.*

Proposition 2 (Logical independence of properties (U1) and (U2)).

Proof on page 32.

2 (a) *Universality of the full response does not imply universality of an induced response.*

2 (b) *Universality of an induced response does not imply universality of the full response.*

Remark 2 (Identical detector simulation). *Processing both generators with the same DELPHES executable and detector card [8] fixes the full response K (up to the caveats of corollary 1), but it does not fix the conditional composition laws $Q_s^{(\cdot)}$. By lemma 2, the same stochastic map averaged against two different composition laws yields two different induced responses. An identical detector simulation is therefore not sufficient for induced universality.*

The same mechanism applies when the nominal “full” representation omits variables on which the detector response depends.

Corollary 1 (Incomplete particle-level representation). *If the detector response is causally determined by an augmented particle-level state $\tilde{T} = (T, J) \in \mathcal{T} \times \mathcal{J}$ through a universal kernel $\tilde{K} : \mathcal{T} \times \mathcal{J} \rightarrow \mathcal{P}(\mathcal{Y})$, where $(\mathcal{J}, \mathcal{B}(\mathcal{J}))$ is a standard Borel space and J is not included in the represented state T , then the represented nominal full response satisfies* Proof on page 32.

$$\forall s \in \mathcal{S} \quad Q_s \left\{ t \in \mathcal{T} \mid K_s(\cdot \mid t) = \int_{\mathcal{J}} \tilde{K}(\cdot \mid t, j) p_s(\mathrm{d}j \mid t) \right\} = 1, \quad (29)$$

where $p_s(\cdot \mid t) \in \mathcal{P}(\mathcal{J})$ is a regular conditional distribution of $J \mid T = t$ under generator $s \in \mathcal{S}$. K_s is therefore an induced response obtained by marginalising over J in the sense of definition 1.

On the common essential support $\mathcal{T}_{\cap}$ under the balanced mixture $\overline{Q} = \frac{1}{2}(Q_H + Q_P)$, disagreement of the conditional distributions,

$$\overline{Q}\{t \in \mathcal{T}_{\cap} \mid p_H(\cdot \mid t) \neq p_P(\cdot \mid t)\} > 0, \quad (30)$$

is necessary for non-universality of K , and is also sufficient if for $\overline{Q}$ -almost every $t \in \mathcal{T}_{\cap}$, the fibre operator $\eta \mapsto \int_{\mathcal{J}} \tilde{K}(\cdot \mid t, j) \eta(\mathrm{d}j)$ on $\mathcal{M}(\mathcal{J})$ is injective.

4 Conditional classification.

4.1 Population quantities.

Definition 5 (Balanced mixture). Let S be a generator indicator with the prior $\mathbb{P}(S = H) = \mathbb{P}(S = P) = \frac{1}{2}$, and let $\mathbb{P}_s$ denote the regular conditional distribution of $(T, Y) \mid S = s$. The resulting joint law $\mathbb{P}$ of $(S, T, Y) \in \mathcal{S} \times \mathcal{T} \times \mathcal{Y}$ is defined to be the *balanced mixture*.

Under $\mathbb{P}$, the marginal law of Z is ν (see remark 1), and $r_s(\cdot | Z)$ is a version of the conditional distribution of $(X | Z, S = s)$.

For a random element W defined on the same space, define

$$\eta_W = \mathbb{P}(S = H | W) \quad (31)$$

$$h(p) = -p \ln p - (1 - p) \ln(1 - p) \quad (32)$$

$$d_{\text{KL}}(p \| q) = p \ln \frac{p}{q} + (1 - p) \ln \frac{1 - p}{1 - q} \quad (33)$$

h is the binary entropy and d_{KL} is the Kullback–Leibler divergence between $\text{Bern}(p)$ and $\text{Bern}(q)$.

All logarithms are base e and all information quantities are in nats.

Remark 3. *Note that*

$$\lim_{a \rightarrow 0} a \ln a = 0 \quad \lim_{a \rightarrow 0} p \ln \frac{p}{a} = +\infty \text{ for } p > 0 \quad (34)$$

Definition 6 (Conditional entropy). The *conditional entropy* of S given W is $H(S | W) = \mathbb{E}[h(\eta_W)]$.

Definition 7 (Conditional mutual information). The *conditional mutual information* between S and X given Z is

$$I(S; X | Z) = \mathbb{E}[d_{\text{KL}}(\eta_{(Z,X)} \| \eta_Z)]. \quad (35)$$

For a binary attribute, eq. (35) coincides with the general definition [9]

$$I(S; X | Z) = \mathbb{E}_Z[D_{\text{KL}}(\mathbb{P}_{(S,X)|Z} \| \mathbb{P}_{S|Z} \otimes \mathbb{P}_{X|Z})]. \quad (36)$$

Proposition 3 (Induced universality as conditional independence). *Under the balanced mixture, the following are equivalent:* Proof on page 32.

3 (a) *The induced response on (f, g) is universal across $\mathcal{S}$;*

3 (b) $X \perp\!\!\!\perp S | Z$;

3 (c) $I(S; X | Z) = 0$.

The magnitude of any violation admits a direct interpretation in terms of the discrepancy between the induced response kernels.

Definition 8 (Skewed JS divergence). For $\alpha \in [0, 1]$, and probability measures $p, q \in \mathcal{P}(\mathcal{X})$, the α -skewed Jensen–Shannon divergence is

$$\text{JS}_\alpha(p, q) = \alpha D_{\text{KL}}[p \parallel \alpha p + (1 - \alpha)q] + (1 - \alpha) D_{\text{KL}}[q \parallel \alpha p + (1 - \alpha)q] \quad (37)$$

Proposition 4 (Conditional mutual information as a Jensen–Shannon divergence).

Proof on
page 33.

$$\begin{aligned} \forall z \in \mathcal{Z} \quad \eta(z) &= \frac{1}{2} m_{\text{H}}(z) \quad \wedge \quad \bar{r}(\cdot \mid z) = \eta(z) r_{\text{H}}(\cdot \mid z) + (1 - \eta(z)) r_{\text{P}}(\cdot \mid z) \\ \implies \eta(Z) &\text{ is a version of the posterior } \eta_Z = \mathbb{P}(S = \text{H} \mid Z) \\ &\wedge I(S; X \mid Z) = \int_{\mathcal{Z} \cap} \text{JS}_{\eta(z)}(r_{\text{H}}(\cdot \mid z), r_{\text{P}}(\cdot \mid z)) \nu(dz). \end{aligned} \quad (38)$$

Proposition 4 shows that $I(S; X \mid Z)$ aggregates the pointwise discrepancy between the two induced responses over the common support, weighted by the local generator balance, and that the non-overlapping regions of the particle-level spectrum do not contribute. Since $\text{JS}_\alpha \leq h(\alpha) \leq \ln 2$, the conditional mutual information is universally bounded by $\ln 2 \approx 0.693 \text{ nat}$.

4.2 Classifier losses.

Definition 9 (Binary log loss and BCE). For a generator label $s \in \mathcal{S}$ and predicted probability $p \in [0, 1]$, the *binary logarithmic loss* $\ell(s, p) \in [0, \infty]$ is

$$\ell(s, p) = -\mathbb{1}_{s=\text{H}} \ln p - \mathbb{1}_{s=\text{P}} \ln(1 - p), \quad (39)$$

continuously extended on the boundaries $p \in \{0, 1\}$ through the conventions of remark 3.

For a feature vector W taking values in a standard Borel space $\mathcal{W}$, and any Borel-measurable classifier $D : \mathcal{W} \rightarrow [0, 1]$, the *population cross-entropy risk* is

$$L_W(D) = \mathbb{E}[\ell\{S, D(W)\}] \in [0, \infty]. \quad (40)$$

Proposition 5 (Log-loss decomposition and Bayes identity). For every measurable classifier $D : \mathcal{W} \rightarrow [0, 1]$,

Proof on
page 34.

$$L_W(D) = H(S \mid W) + \mathbb{E}[d_{\text{KL}}\{\eta_W \parallel D(W)\}]. \quad (41)$$

Consequently, $L_W(D) \geq H(S \mid W)$, with equality iff $D(W) = \eta_W$ almost surely. Furthermore, the Bayes-optimal classifiers $D_Z^*(Z) = \eta_Z$ and $D_{ZX}^*(Z, X) = \eta_{(Z, X)}$ attain the respective conditional entropies, yielding the identity

$$L_Z(D_Z^*) - L_{ZX}(D_{ZX}^*) = H(S \mid Z) - H(S \mid Z, X) = I(S; X \mid Z). \quad (42)$$

Equation (41) is the statement that logarithmic loss is a strictly proper scoring rule [10]. The risk difference in eq. (42) establishes that the conditional mutual information $I(S; X | Z)$, and hence the severity of response non-universality, is precisely the reduction in Bayes risk achieved by adding the detector observable X to a classifier already trained on the particle summary Z . Hence same decomposition controls the behaviour of learned classifiers.

Corollary 2 (Learned classifiers). *For measurable D_Z and D_{ZX} with finite losses,* Proof on
page 35.

$$\left\{ \begin{array}{l} \varepsilon_Z = L_Z(D_Z) - H(S | Z) \geq 0 \\ \wedge \varepsilon_{ZX} = L_{ZX}(D_{ZX}) - H(S | Z, X) \geq 0 \\ \wedge \Delta = L_Z(D_Z) - L_{ZX}(D_{ZX}) \end{array} \right\} \implies \Delta = I(S; X | Z) + \varepsilon_Z - \varepsilon_{ZX} \quad (43)$$

In particular:

- 2(a) $I(S; X | Z) = 0 \implies \Delta = \varepsilon_Z - \varepsilon_{ZX}$, which is positive whenever the classifier using Z alone is fitted less well than the classifier using (Z, X) ;
- 2(b) $\varepsilon_Z \leq \varepsilon_{ZX} \implies \Delta \leq I(S; X | Z)$, so that $\Delta > 0 \implies I(S; X | Z) > 0$;
- 2(c) in all cases, $\Delta \leq I(S; X | Z) + \varepsilon_Z$.

Corollary 2 (a) identifies the relevant failure mode of the test. A positive gain under the null hypothesis requires the lower-dimensional classifier to be the worse-fitted of the two. Held-out evaluation protects against overfitting of the training set but not against this asymmetry, which may arise from differences in approximation, optimisation, or calibration error. Corollary 2 (b) shows that, conversely, the gain is a conservative estimate of $I(S; X | Z)$ whenever the lower-dimensional problem is fitted at least as well.

4.3 Estimator.

Definition 10 (Paired, held out estimator). Let $\hat{D}_Z$ and $\hat{D}_{ZX}$ be classifiers fitted on training and validation data, and let $\{(S_i, Z_i, X_i)\}_{i=1}^n$ be an independent test sample with $n_H = n_P = \frac{n}{2}$ events drawn independently from each generator. The *paired held-out estimator* is

$$\hat{\delta} = \frac{1}{n} \sum_{i=1}^n d_i, \quad d_i = \ell(S_i, \hat{D}_Z(Z_i)) - \ell(S_i, \hat{D}_{ZX}(Z_i, X_i)). \quad (44)$$

Scoring both classifiers on the same events removes the component of event-to-event variance that is common to the two losses.

Proposition 6 (Conditional sampling properties). *Conditional on the fitted classifiers $\hat{D}_Z$ and $\hat{D}_{ZX}$,* Proof on page 35.

$$\begin{aligned} \forall s \in \mathcal{S} \quad \sigma_s^2 &= \text{Var}(d_i \mid S_i = s) < \infty \\ \implies \mathbb{E}[\hat{\delta} \mid \hat{D}_Z, \hat{D}_{ZX}] &= \Delta(\hat{D}_Z, \hat{D}_{ZX}), \quad \text{Var}(\hat{\delta} \mid \hat{D}_Z, \hat{D}_{ZX}) = \frac{\sigma_H^2 + \sigma_P^2}{2n}, \end{aligned} \quad (45)$$

where $\Delta(\cdot, \cdot)$ is the population gain of corollary 2 under the balanced mixture.

The standard error reported below is the plug-in estimate

$$\widehat{\text{SE}}_{\text{event}} = \sqrt{\frac{\widehat{\text{Var}}(d_i \mid S_i = \text{H}) + \widehat{\text{Var}}(d_i \mid S_i = \text{P})}{2n}}. \quad (46)$$

By construction, it quantifies test-event sampling variation conditional on the fitted classifiers only. It does not include variation due to the training sample, random initialisation, hyperparameter selection, or multiplicity of analyses, and it does not account for the excess-risk asymmetry of corollary 2. The ratio $\frac{\widehat{\delta}}{\widehat{\text{SE}}_{\text{event}}}$ is therefore reported as a scale diagnostic and not as a calibrated significance.

5 Empirical study.

5.1 Samples and implementation.

The study uses the HERWIG+DELPHES and PYTHIA+DELPHES samples of Andreassen et al. [1] (see section 7 for generator and detector versions). The scalar observables are the six jet-substructure observables considered there: the ungroomed jet mass m , the constituent multiplicity M , the jet width w , the logarithm of the soft-drop groomed jet mass normalised by the jet transverse momentum $\ln \rho$ [11], the N -subjettiness ratio τ_{21} [12], and the soft-drop momentum fraction z_g [11]. For each observable, the classification data are the triples

$$(Z, X, S) = \begin{cases} (z_{\text{truth}}, x_{\text{data}}, \text{H}), & \text{HERWIG} + \text{DELPHES}, \\ (z_{\text{gen}}, x_{\text{sim}}, \text{P}), & \text{PYTHIA} + \text{DELPHES}. \end{cases} \quad (47)$$

Table 1: Held-out conditional-classification results for the six scalar observables. The columns give the paired held-out gain in binary cross-entropy from including the detector-level observable (see eq. (44)), the conditional test-event standard error (see eq. (46)), and their signal-to-noise ratio.

Observable	$\hat{\delta}$ [mnat/event]	$\widehat{\text{SE}}_{\text{event}}$ [mnat/event]	$\hat{\delta}/\widehat{\text{SE}}_{\text{event}}$
m	9.061	0.465	19.5
M	2.142	0.255	8.4
w	0.562	0.122	4.6
τ_{21}	1.102	0.161	6.8
z_g	1.216	0.192	6.3
$\ln \rho$	1.897	0.232	8.2

Particle- and detector-level values remain paired event by event within each generator.

Each run uses 200 000 events per generator. Particle- and detector-level values are standardised using the mean and standard deviation of z_{gen} . The events are divided into fixed training, validation and test partitions, and the two generators are balanced exactly within each partition. Two classifiers, $\hat{D}_Z$ and $\hat{D}_{ZX}$, are trained separately. Each is a fully connected network with three hidden layers of 128 units, trained with the ADAM optimiser [13] at learning rate 10^{-3} and batch size 1024 for at most 200 epochs, with early stopping at a patience of 10 epochs. Both classifiers are evaluated on the same balanced test events, and $\hat{\delta}$ and $\widehat{\text{SE}}_{\text{event}}$ are computed according to eqs. (44) and (46).

5.2 Results.

The empirical findings are summarized in table 1. All six observables yield a strictly positive empirical gain $\hat{\delta}$, spanning 4.6 to 19.5 conditional standard errors above zero. The gain for the ungroomed jet mass m (9.061 mnat/event) is particularly pronounced, exceeding the second-largest gain (M) by a factor of 4.2 and the smallest (w) by a factor of 16.1; the remaining five observables fall within 0.562 mnat/event to 2.142 mnat/event.

For the joint six-dimensional summary vector $(\mathbf{Z}, \mathbf{X}) \in \mathbb{R}^6 \times \mathbb{R}^6$, an identical evaluation protocol yields $\hat{\delta} = (3.559 \pm 0.433)$ mnat/event, where the parenthetical uncertainty denotes $\widehat{\text{SE}}_{\text{event}}$. The joint estimate is notably smaller than the sum of

the univariate gains. In fact, the joint gain is smaller than the mass gain alone. This is consistent with the composition hypothesis, since the other truth observables may absorb part of the mass residual. Such non-additivity is expected: by the chain rule for conditional mutual information, $I(S; \mathbf{X} \mid \mathbf{Z})$ need not be equal to $\sum_k I(S; X_k \mid Z_k)$ because the conditioning particle summary $\mathbf{Z}$ differs from each individual coordinate Z_k , the substructure variables exhibit mutual statistical redundancy, and higher dimensional classifiers face different capacity and optimization trade-offs.

5.3 Constructed-null diagnostic.

To probe the excess-risk asymmetry $\varepsilon_Z - \varepsilon_{ZX}$ of corollary 2 (a), construct synthetic pseudo-domains that satisfy the conditional-independence null hypothesis exactly. Within a single generator sample $s \in \mathcal{S}$, define a measurable propensity function $\pi: \mathcal{Z} \rightarrow (0, 1)$ and assign a binary pseudo-label $S' \in \{0, 1\}$ conditionally via $\mathbb{P}(S' = 1 \mid Z, X) = \pi(Z)$. By construction, $X \perp\!\!\!\perp S' \mid Z$, ensuring $I(S'; X \mid Z) = 0$, while the marginal distributions of Z across the two pseudo-classes differ whenever π is non-constant. Balancing the pseudo-domains by uniform rejection subsampling preserves the conditional law of X given (Z, S') , thereby maintaining the null property. Evaluating the full training and estimation pipeline on these null samples (with half the per-class sample size of the primary analysis to allow balanced class splitting) yields an average gain over five independent runs of $\hat{\delta} = (-0.329 \pm 0.777)$ mnat/event within HERWIG and (-0.458 ± 0.328) mnat/event within PYTHIA, where uncertainties denote across run standard deviations.

Under the null, corollary 2 (a) dictates $\Delta = \varepsilon_Z - \varepsilon_{ZX}$. The corresponding standard errors of the mean are 0.347 mnat/event and 0.147 mnat/event, so the HERWIG mean is consistent with zero, whereas the PYTHIA mean lies approximately 3.1 standard errors below it. The sign indicates that the higher dimensional joint model D_{ZX} tends to incur a small optimization or generalization penalty ($\varepsilon_{ZX} \geq \varepsilon_Z$) rather than an artificial advantage, so that the test is, if anything, conservative. The run-to-run standard deviations are the appropriate scale against which to compare the single-run gains of table 1. The observed gain for m exceeds the null run-to-run standard deviation by more than an order of magnitude for either generator, and those for M and $\ln \rho$ by more than a factor of two; the gains for z_g and τ_{21} are distinguishable only relative to the smaller PYTHIA spread, and that for w is not distinguishable from the null. Because this pseudo-domain construction operates at a reduced sample size with synthetic marginal separation, it is intended as an architectural diagnostic rather than a formal permutation calibrated hypothesis test for computing p -values.

5.4 Interpretation.

By proposition 3, the conditional-independence hypothesis $I(S; X_m | Z_m) = 0$ is mathematically equivalent to universality of the induced response for the ungroomed jet mass m . Under this null hypothesis, corollary 2 (a) and proposition 6 dictate that the observed empirical gain of 9.061 mnat/event would have to originate entirely from an excess-risk asymmetry $\varepsilon_Z - \varepsilon_{ZX}$ of identical magnitude, modulo test-sample fluctuations on the order of 0.5 mnat/event. However, the synthetic null benchmark yields mean asymmetries that are more than an order of magnitude smaller and negative, with run-to-run standard deviations below 1 mnat/event. Table 1 can therefore be regarded as strong, though not formally calibrated, evidence that the induced mass response is not universal between HERWIG and PYTHIA, and that among the tested observables this violation is by far the largest for ungroomed jet mass. Only the statement that m carries by far the largest effect is robust to training variability; the ordering among the remaining five observables is not resolved at this sample size.

Remark 4 (Scope of the conclusion). *Non-universality of the induced response rules out the existence of any single transition kernel $r : \mathcal{Z} \rightarrow \mathcal{P}(\mathcal{X})$ that simultaneously factorizes the joint distribution of (Z, X) via $\mu_s(dz) r(dx | z)$ for both generators $s \in \mathcal{S}$. Crucially, this does not rule out the existence of a kernel r that successfully matches both detector-level marginal distributions,*

$$\forall s \in \mathcal{S} \quad \int_{\mathcal{Z}} r(\cdot | z) \mu_s(dz) = g_{\#}(K_s Q_s) \quad (48)$$

which is a strictly weaker consistency condition. Finally, as established by proposition 2 (a), non-universality of the induced response r does not imply failure of universality at the underlying microscopic K .

6 Consequences for unfolding.

6.1 Unfolding with a misspecified induced response.

Consider unfolding performed directly in the summary space $\mathcal{Z}$. By eq. (2), the law of the detector-level summary in nature is

$$\mathbb{P}_{\text{data}} = \mathbb{P}_{\text{H}}(X \in \cdot) = r_{\text{H}} \mu_{\text{H}}, \quad (49)$$

which holds without any modelling assumption. An analysis that transfers the PYTHIA induced response instead uses the forward model

$$\mu \mapsto r_{\text{P}} \mu, \quad \mu \in \mathcal{P}_{\text{P}} = \{\mu \in \mathcal{P}(\mathcal{Z}) \mid \mu \ll \mu_{\text{P}}\}, \quad (50)$$

where absolute continuity with respect to μ_P ensures that $r_P\mu$ does not depend on the choice of version of r_P .

Definition 11 (Non-Cancellation). The *non-cancellation condition* is the strict inequality between the transferred forward fold of the true summary and the true detector law:

$$r_P\mu_H \neq r_H\mu_H. \quad (51)$$

Proposition 7 (Closure and truth are incompatible under a misspecified induced response). *If $\mu_H \ll \mu_P$, $r_P\mu_H \neq r_H\mu_H$, and $\Theta_{cl} = \{\mu \in \mathcal{P}_P \mid r_P\mu = \mathbb{P}_{data}\}$ is the set of particle-level distributions achieving exact detector-level closure under the transferred response, then:* Proof on [page 35](#).

- 7 (a) $\mu_H \notin \Theta_{cl}$; that is, the true particle-level distribution, folded through the transferred response, does not reproduce the detector-level data;
- 7 (b) if $\Theta_{cl} \neq \emptyset$, then $\Theta_{cl} = \operatorname{argmin}_{\mu \in \mathcal{P}_P} D_{KL}(\mathbb{P}_{data} \parallel r_P\mu)$, so that every population maximum-likelihood solution differs from μ_H .

Thus, for summary-level unfolding, even an exact optimiser of the misspecified forward model faces a forced choice: it can reproduce the true particle-level distribution or achieve detector-level closure, but not both.

Remark 5 (Folded mismatch versus induced non-universality). Equation (51) is strictly stronger than non-universality of the induced response, since the signed difference measure,

$$r_P\mu_H - r_H\mu_H = \int_{\mathcal{Z}} [r_P(\cdot \mid z) - r_H(\cdot \mid z)] \mu_H(dz) \quad (52)$$

can vanish through cancellation across z even when the integrand does not vanish μ_H -almost everywhere. The conditional classification test of section 4 detects pointwise discrepancies in the integrand, whereas integrated mismatch can be probed directly via closure tests with known truth (see section 8.3).

6.2 Non-identifiability under a universal full response.

The failure described above arises from a reduced and misspecified response. A second, logically distinct failure persists even when the full response is exactly universal and exactly known.

Throughout this subsection, let:

- The full response be universal across $\mathcal{S}$ with a known common version $K : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{Y})$.
- $\mathcal{Q} \subseteq \mathcal{P}(\mathcal{T})$ be a statistical model of particle-level distributions.
- The detector-level observations $\{Y_i\}_{i=1}^n$ be independent and identically distributed with law KQ for some $Q \in \mathcal{Q}$.

The target of inference is the pushforward distribution $f_{\#}Q$ of the particle-level summary.

Definition 12 (Estimator). An *estimator* is a measurable map $\hat{\phi}_n : \mathcal{Y}^n \rightarrow \mathcal{P}(\mathcal{Z})$, where $\mathcal{P}(\mathcal{Z})$ is equipped with a metric d that metrises weak convergence, such as the bounded-Lipschitz metric.

Theorem 2 (Non-identifiability under a universal full response). *Let $K : \mathcal{T} \rightarrow \mathcal{P}(\mathcal{Y})$ be a universal response kernel, and let $\mathcal{Q} \subseteq \mathcal{P}(\mathcal{T})$ be a model class of particle-level distributions. If* Proof on [page 36](#).

$$\exists Q_1, Q_2 \in \mathcal{Q} \quad KQ_1 = KQ_2 \quad \wedge \quad f_{\#}Q_1 \neq f_{\#}Q_2, \quad (53)$$

then:

- 2 (a) For any sequence of measurable estimators $\hat{\phi}_n : \mathcal{Y}^n \rightarrow \mathcal{P}(\mathcal{Z})$, $\{\hat{\phi}_n\}_{n \in \mathbb{N}}$ is not consistent for $f_{\#}Q$ simultaneously at Q_1 and Q_2 ; that is,

$$\limsup_{n \rightarrow \infty} \max_{i \in \{1,2\}} \mathbb{P}_{Q_i} \left(d(\hat{\phi}_n, f_{\#}Q_i) \geq \frac{1}{2} d(f_{\#}Q_1, f_{\#}Q_2) \right) \geq \frac{1}{2}, \quad (54)$$

where d is any metric on $\mathcal{P}(\mathcal{Z})$.

- 2 (b) For detector-level data generated under Q_1 ,

$$Q_2 \in \operatorname{argmin}_{Q \in \mathcal{Q}} D_{\text{KL}}(KQ_1 \parallel KQ) = \{Q \in \mathcal{Q} \mid KQ = KQ_1\}, \quad (55)$$

so that the set of population maximum-likelihood solutions contains distributions whose pushforwards strictly differ from $f_{\#}Q_1$.

The difference $Q_1 - Q_2 \in \ker K$ is a *null direction* of the full detector response. Along a null direction, the observation likelihood is strictly invariant, and the specific distribution returned by an unfolding procedure is determined entirely by inductive bias, such as its prior, regularisation functional, network architecture, parameter

initialisation, or early-stopping criterion, rather than by the empirical data. This is the precise sense in which a result is *prior dependent*. Exact null directions represent an idealisation; the following quantitative lower bound governs approximate null directions.

Proposition 8 (Near null directions).

Proof on
page 36.

$$\forall Q_1, Q_2 \in \mathcal{Q} \quad \forall n \in \mathbb{N} \quad \forall \hat{\phi}_n: \mathcal{Y}^n \rightarrow \mathcal{P}(\mathcal{Z})$$

$$\max_{i \in \{1,2\}} \mathbb{P}_{Q_i} \left[d(\hat{\phi}_n, f_{\#} Q_i) \geq \frac{1}{2} d(f_{\#} Q_1, f_{\#} Q_2) \right] \geq \frac{1}{2} - \frac{n}{2} d_{\text{TV}}(K Q_1, K Q_2). \quad (56)$$

Hence particle-level differences along directions with $d_{\text{TV}}(K Q_1, K Q_2) \ll \frac{1}{n}$ are unresolvable by any method at sample size n , including an oracle unfolding with exact knowledge of K .

6.2.1 Physical realisation.

At particle-level, the invariant mass of a jet depends on the 4-momenta of all constituent particles, which in turn incorporate the rest masses tied to specific hadron species. At detector-level, particle-flow reconstruction assigns reconstructed objects to broad experimental categories (e.g., charged hadrons, neutral hadrons, photons) with nominal mass hypotheses, losing fine-grained particle identification. Consequently, distinct particle-level states with differing species compositions, and therefore differing true invariant masses, can produce detector signatures that are nearly or strictly indistinguishable.

Example 1 is an idealized model of this physical mechanism. By lemma 3 (d), the generator distributions Q_H and Q_P realize the exact null direction condition of eq. (53). While an unbinned, high dimensional unfolding method can leverage subtle kinematic correlations elsewhere in the event to constrain the species mixture, it cannot reconstruct information that the detector response has fundamentally erased. Thus, although unfolding in the full phase space avoids the information loss incurred by restricting to a low dimensional summary, it cannot overcome the intrinsic null directions of the detector response operator itself.

7 Application to the OMNIFOLD jet-mass result.

Andreassen et al. [2] uses HERWIG 7.1.5 [14] as “Nature” and PYTHIA 8.243 [15] with Tune 26 as MC. DELPHES 3.4.2 with a CMS detector model [8] serves as the detector

Table 2: Table I of Andreassen et al. [2], reproduced. Entries are $\Delta_{VLC} \times 10^3$ between unfolded and particle-level binned histograms. “Data” denotes the uncorrected detector-level distribution and “Generation” the PYTHIA prior.

Method	m	M	w	$\ln \rho$	τ_{21}	z_g
OMNIFOLD	2.77	0.33	0.10	0.35	0.53	0.68
MULTIFOLD	3.80	0.89	0.09	0.37	0.26	0.15
UNIFOLD	8.82	1.46	0.15	0.59	1.11	0.59
IBU	9.31	1.51	0.11	0.71	1.10	0.37
Data	24.6	130	15.7	14.2	11.1	3.76
Generation	3.62	15	22.4	19	20.8	3.84

simulation. OMNIFOLD is implemented with Particle Flow Networks [16] acting on sets of particles described by transverse momentum, rapidity, azimuth and a particle identification code, where at detector-level the identification is restricted to experimentally accessible categories. The likelihood derivation of Andreassen et al. [2] assumes

$$p_{\text{Data}|\text{Truth}}(y | t) = p_{\text{Sim}|\text{Gen}}(y | t), \quad (57)$$

which, in the framework of section 2, posits exact universality of the full response kernel across the represented state spaces $\mathcal{T}$ and $\mathcal{Y}$.

Table 2 is a reproduction of the paper’s Table I, which reports values of the triangular discriminator [17],

$$\Delta_{VLC}(p, q) = \frac{1}{2} \int \frac{(p(\lambda) - q(\lambda))^2}{p(\lambda) + q(\lambda)} d\lambda \quad (58)$$

between unfolded and particle-level histograms.

Three features of table 2 are particularly notable:

1. OMNIFOLD achieves its poorest performance on jet mass. The residual metric of 2.77×10^{-3} is approximately four times larger than the next-highest entry (0.68×10^{-3} , for z_g) and between five and thirty times larger than each of the remaining observables.
2. The mass discrepancy contracts monotonically as richer event information is included in the deconvolution, decreasing from 8.82×10^{-3} (UNIFOLD) to 3.80×10^{-3} (MULTIFOLD) and 2.77×10^{-3} (OMNIFOLD), yet it remains away from zero.

3. Jet mass is the sole observable for which low-dimensional deconvolution (UNIFOLD and IBU) degrades performance relative to the unweighted prior (3.62×10^{-3}); MULTIFOLD likewise fails to improve upon the prior, leaving OMNIFOLD as the only method to achieve a net reduction.

The third feature aligns directly with the distortion predicted by proposition 7 when summary-level unfolding is conducted under a misspecified induced response, although table 2 alone cannot isolate that mechanism uniquely. The second feature is consistent with higher-dimensional architectures capturing indirect composition signatures from the broader shower, with the residual gap reflecting information either fundamentally erased by detector smearing (see theorem 2) or distorted by cross-generator response misspecification.

Andreassen et al. [2] attributes the difficulty with mass to particle type information being relevant at particle-level but not fully known at detector-level, introducing additional prior dependence. The following propositions disentangle the distinct physical and statistical mechanisms underlying this observation.

Proposition 9 (Compatibility of the two explanations). *There exists a data generating process in which a single latent particle-level variable simultaneously induces:*

Proof on
page 37.

- 9 (a) *an exactly universal full response kernel K ;*
- 9 (b) *a non-universal induced mass response $r_H(\cdot | z) \neq r_P(\cdot | z)$ at every mass value $z \in \mathcal{Z}$; and*
- 9 (c) *severe non-identifiability of the particle-level mass distribution under the known full response, where Q_H and Q_P form an exact null pair satisfying $KQ_H = KQ_P$ while $f_{\#}Q_H \neq f_{\#}Q_P$.*

Consequently, the particle-identity hypothesis of Andreassen et al. [2] and the non-universality of the induced mass response demonstrated in section 5.4 are mathematically consistent and can originate from the identical latent degree of freedom.

Proposition 9 implies that the non-universality detected in section 5.4 is not, *ipso facto*, incompatible with the full-response universality assumption in eq. (57). OMNIFOLD does not presuppose agreement between the reduced kernels r_H and r_P ; it relies only on universality conditional on the full represented state T , which by proposition 2 (a) can hold even when the induced summary response breaks down. Conversely, however, the empirical validation presented in Andreassen et al. [2] is insufficient to prove that eq. (57) holds, for two distinct reasons.

First, the operational “full phase space” consists strictly of the represented constituent list, which may not capture every latent coordinate that influences DELPHES detector simulation, event selection, or jet-clustering thresholds. If unrepresented response-relevant coordinates affect the simulation, corollary 1 implies that the nominal response is itself an induced marginalization that can inherit generator dependence.

Second, the technical closure test of Andreassen et al. [2] is performed within a single generator, and such a test provides zero statistical power against cross-generator response misspecification.

Proposition 10 (Within-generator closure tests). *If Φ is any test statistic evaluated exclusively on samples drawn from $\mathbb{P}_P$, potentially augmented by independent auxiliary simulation randomness, then the sampling distribution of Φ is identical under any two joint specifications $(\mathbb{P}_H, \mathbb{P}_P)$ and $(\mathbb{P}'_H, \mathbb{P}_P)$ sharing the identical PYTHIA law $\mathbb{P}_P$.*

Proof on
page 37.

In particular, for testing the null hypothesis of response universality $H_0: K_H = K_P$ against any alternative specification $(\mathbb{P}'_H, \mathbb{P}_P)$ under which $K'_H \neq K_P$, any test based on Φ has power identically equal to its size.

The numerical evidence of Andreassen et al. [2] therefore cannot and does not discriminate among at least three sources of the unfolded mass residual:

- (S1) intrinsic non-identifiability or near-null directions under an exactly universal full response (see theorem 2 and proposition 8);
- (S2) cross-generator response misspecification conditional on the represented state T , including discrepancies induced by omitted phase-space coordinates (see corollary 1);
- (S3) finite-sample variance, neural-network approximation error, optimization imperfections, or premature early stopping.

The particle–identity hypothesis provides a coherent explanation for source (S1) and for the observed summary–level non-universality (proposition 9). While within-generator technical closure tests provide checks on source (S3), proposition 10 proves that they cannot constrain source (S2). Attributing the mass discrepancy solely to prior dependence is therefore plausible, but not an empirically verified conclusion.

8 Proposed tests.

The following tests would localise the source of the mass residual.

8.1 Composition augmented conditioning.

Let $C = c(T)$ be a vector of particle-level composition variables, such as charged and neutral energy fractions, constituent multiplicities, species fractions, and species-resolved contributions to the jet mass. Repeat the nested classification test of section 4 with the particle-level summary (Z, C) in place of Z , that is, using classifiers $D_{Z,C}(Z, C)$ and $D_{Z,C,X}(Z, C, X)$.

Corollary 3 (Composition-augmented test). *If the g -projected full response is universal with common version K^g and if $K^g(\cdot | t)$ depends on t only through the augmented coordinates $(f(t), c(t))$, in the sense of eq. (22) with f replaced by the Borel map $(f, c) : \mathcal{T} \rightarrow \mathcal{Z} \times \mathcal{C}$, then* Proof on page 37.

$$I(S; X | Z, C) = 0. \quad (59)$$

If $I(S; X | Z, C_k)$ contracts monotonically towards zero along a sequence of increasingly expressive composition summaries $C_1, C_2, \dots$, this provides direct empirical evidence that omitted particle composition is the latent variable driving the non-universal induced mass response. However, conditional mutual information is generally not monotone under refinement of the conditioning algebra [18]; hence, an intermediate increase along the sequence is mathematically permissible, while a vanishing terminal value remains consistent with, but does not strictly imply, universality of the underlying full response (see proposition 2 (b)).

8.2 Full-state conditional response test.

Using the exact particle- and detector-level representations supplied to OMNIFOLD, consider the population target

$$I(S; Y | T) = L_T(D_T^*) - L_{TY}(D_{TY}^*). \quad (60)$$

Corollary 4 (Full-state test).

Proof on page 37.

4(a) $I(S; Y | T) = 0$ iff the full response is universal across $\mathcal{S}$.

4(b) $I(S; X | T) = 0$ iff the g -projected full response is universal across $\mathcal{S}$. Moreover, because $X = g(Y)$, the data processing inequality guarantees

$$I(S; X | T) \leq I(S; Y | T). \quad (61)$$

A reproducibly positive estimate of $I(S; Y \mid T)$ would formally reject universality of the full response for the represented state and directly invalidate the foundation in eq. (57). Conversely, an estimate consistent with a calibrated null would corroborate the hypothesis that the scalar discrepancy identified in section 5 is strictly a coarse-graining artifact. Corollary 4 (b) establishes an intermediate diagnostic by conditioning on the full particle-level state while probing only the scalar detector-level mass X_m . Combined with lemma 2, this delivers an unambiguous dichotomy:

- If $I(S; X_m \mid T) > 0$, the represented full response itself violates universality in the mass coordinate.
- If $I(S; X_m \mid T) = 0$, the g -projected full response is strictly universal, and the non-universality observed in the induced mass response arises solely from differences in generator conditional composition laws, as dictated by eq. (18).

These high dimensional formulations are substantially more demanding than scalar evaluations for three technical reasons:

1. Variable length particle sets necessitate permutation invariant architectures (such as Particle Flow Networks [16]);
2. The network pairs D_T and D_{TY} must be carefully capacity matched to prevent the excess risk asymmetry $\varepsilon_T - \varepsilon_{TY}$ of corollary 2 from dominating the estimate;
3. Failing to detect conditional dependence cannot be construed as affirmative evidence of conditional independence, since no conditional independence test can be uniformly valid and non-trivially powerful across general continuous alternatives without restricting smoothness or distribution classes [19].

Consequently, power analyses with synthetic response perturbations of known amplitude are necessary before any observed null result can be definitively claimed.

8.3 Cross-generator oracle closure.

Because synthetic truth is accessible in Monte Carlo benchmarks, deconvolution can be evaluated across the complete Cartesian product of detector-level data sources and response-simulation pairs:

	HERWIG response pairs	PYTHIA response pairs
HERWIG detector data	Matched	Transferred
PYTHIA detector data	Transferred	Matched

The matched diagonal cells isolate pure deconvolution algorithmic artifacts and intrinsic non-identifiability (see sources (S1) and (S3)); the discrepancy between off-diagonal transferred cells and matched cells isolates the empirical cost of cross-generator response misspecification (see source (S2)). At the summary-level, the folded mismatch $r_P\mu_H - r_H\mu_H$ (see remark 5) can be evaluated directly by reweighting PYTHIA response pairs to the HERWIG particle-level distribution of Z and comparing the resulting folded distribution against the nominal HERWIG detector-level observations. Potential differences in generator support ($\mu_H \not\ll \mu_P$ or the converse) must be audited to prevent truncation bias.

8.4 Calibration and uncertainty quantification.

To convert the scale diagnostic of section 5 into a calibrated hypothesis test with a verified power envelope, the following protocol should be implemented:

1. Replicate the entire training, validation, and scoring sequence across independent Monte Carlo resamples and network parameter initialisations;
2. Implement K -fold cross-fitting [20] to ensure that every evaluation event is strictly out of sample;
3. Calibrate empirical test statistics against synthetic conditional nulls or model- X conditional randomisation tests [21] matched in sample size to the observed problem;
4. Construct empirical learning curves across sample sizes and network parameter scales to establish empirical bounds on the excess risks of corollary 2;
5. Validate the probability calibration of network outputs (or evaluate strictly proper scoring rules across model ensembles);
6. Lock the test partition until all neural network architectures, optimisation schedules, and hyperparameters are irrevocably finalised.

9 Conclusions.

The presented benchmark provides strong empirical evidence that the response induced on ungroomed jet mass is not universal between HERWIG and PYTHIA, and weaker but consistent evidence for M , $\ln \rho$, z_g and τ_{21} , despite the use of an identical DELPHES detector simulation. Ungroomed jet mass is singled out among the tested observables because at fixed particle-level mass, the detector-level mass retains several times more residual generator information than the detector-level counterpart of any other substructure observable.

For unfolding performed directly in a reduced summary space, this lack of universality constitutes fundamental model misspecification. Whenever the folded mismatch of eq. (51) is non-zero, no estimator operating under the transferred response can simultaneously recover the true particle-level distribution and reproduce the detector-level data (see proposition 7). In this precise sense, no optimisation scheme or regularisation strategy can rescue summary-level unfolding under response transfer.

For full phase-space unfolding via OMNIFOLD, the implications are more nuanced. Even an exactly universal full response may admit non-trivial null directions, and the detector-level erasure of particle identity can render the particle-level mass distribution fundamentally prior dependent even when the response operator is known exactly (see theorem 2 and proposition 8). At the same time, neither the scalar findings of table 1 nor within-generator technical closure tests can establish whether the represented full response is truly shared between HERWIG and PYTHIA (see propositions 2 and 10). The particle identity explanation of Andreassen et al. [2] is therefore plausible and mathematically consistent with the results established here (see proposition 9), but it remains an unverified hypothesis.

The central open question is whether the independence condition

$$Y \perp\!\!\!\perp S \mid T \tag{62}$$

holds for the full represented particle- and detector-level event states under the balanced mixture. By corollary 4, evaluating this hypothesis directly on full paired particle sets will decisively distinguish a genuine breakdown of the foundational response assumption in eq. (57) from irreducible prior dependence under a universal but non-injective response operator.

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A Proofs.

Throughout, it is understood that equalities between kernels are up to null sets of the relevant marginal and that two finite measures on a product of standard Borel spaces that agree on measurable rectangles are equal, since rectangles form a π -system generating the product σ -algebra.

Proof of Lemma 2. By universality of the g -projected full response and the definition of the fibre disintegration in eq. (3),

$$\forall s \in \mathcal{S} \quad \forall A \in \mathcal{B}(\mathcal{X}) \quad \forall B \in \mathcal{B}(\mathcal{Z})$$

$$\mathbb{P}_s(Z \in B, X \in A) = \int_{f^{-1}(B)} K^g(A | t) Q_s(dt) = \int_B \left[\int_{\mathcal{T}} K^g(A | t) Q_s^z(dt) \right] \mu_s(dz). \quad (63)$$

The second equality follows because $\mathbb{1}_{f^{-1}(B)}(t) = \mathbb{1}_B(f(t))$, and eq. (3) extends by standard monotone-class arguments to

$$\mathbb{E}_s[\psi(Z, T)] = \int_{\mathcal{Z}} \int_{\mathcal{T}} \psi(z, t) Q_s^z(dt) \mu_s(dz), \quad (64)$$

for any bounded Borel-measurable function ψ , applied here to $\psi(z, t) = \mathbb{1}_B(z) K^g(A | t)$.

The assignment $z \mapsto \int_{\mathcal{T}} K^g(\cdot | t) Q_s^z(dt)$ defines a composition of Markov kernels and is therefore itself a Markov kernel $\mathcal{Z} \rightarrow \mathcal{X}$. Comparing this with the definition of the induced response in eq. (2), the two kernels assign identical probabilities to all measurable rectangles $B \times A$, and hence define the same joint measure on $\mathcal{Z} \times \mathcal{X}$. By the uniqueness statement of lemma 1, they are versions of one another. For eq. (17), identify $\mathcal{T}$ with $\mathcal{Z} \times \mathcal{U}$; then $Q_s^z = \delta_z \otimes p_s(\cdot | z)$ for μ_s -almost every z , and eq. (16) reduces to eq. (17). $\square$

Proof of Theorem 1. By lemma 2,

$$\forall s \in \mathcal{S} \quad \exists N_s \in \mathcal{B}(\mathcal{Z}) \quad \mu_s(N_s) = 0 \quad \wedge \quad \forall z \in \mathcal{Z} \setminus N_s \quad r_s(\cdot | z) = K^g Q_s^z. \quad (65)$$

On the common essential support $\mathcal{Z}_\cap$, the Radon–Nikodym derivatives $m_s = \frac{d\mu_s}{d\nu} > 0$ by definition, so $\forall s \in \mathcal{S} \quad \nu \upharpoonright_{\mathcal{Z}_\cap} \ll \mu_s$.

Consequently, $\nu(N_s \cap \mathcal{Z}_\cap) = 0$, and both representations hold simultaneously for ν -almost every $z \in \mathcal{Z}_\cap$. Subtracting the two instances and using linearity of K^g on $\mathcal{M}(\mathcal{T})$ (see definition 3) yields eq. (18).

Theorem 1 (a) By remark 1:

$$\text{The induced response is universal} \iff \nu\{z \in \mathcal{Z}_\cap \mid r_H(\cdot | z) - r_P(\cdot | z) = 0\} = 1. \quad (66)$$

By eq. (18), this is equivalent to $K^g(Q_H^z - Q_P^z) = 0 \implies Q_H^z - Q_P^z \in \ker K^g$.

Theorem 1 (b) If $Q_H^z = Q_P^z$, their difference is the zero measure, which is in $\ker K^g$ trivially; hence the claim follows from theorem 1 (a).

Theorem 1 (c) By lemma 1, for ν -almost every $z \in \mathcal{Z} \cap Q_H^z, Q_P^z$ are concentrated on the fibre $f^{-1}\{z\} \implies Q_H^z - Q_P^z \in \mathcal{M}(f^{-1}\{z\})$. Under the injectivity hypothesis on this subspace, $K^g(Q_H^z - Q_P^z) = 0 \iff Q_H^z - Q_P^z = 0$. The assertion then follows directly from theorem 1 (a). $\square$

Proof of Proposition 1.

$$\begin{aligned} \forall A \in \mathcal{B}(\mathcal{X}) \quad \forall B \in \mathcal{B}(\mathcal{Z}) \quad \forall s \in \mathcal{S} \quad \mathbb{P}_s(Z \in B, X \in A) \\ = \int_{f^{-1}(B)} K^g(A \mid t) Q_s(dt) = \int_{f^{-1}(B)} \kappa(A \mid f(t)) Q_s(dt) = \int_B \kappa(A \mid z) \mu_s(dz), \end{aligned} \quad (67)$$

where the first equality is the definition of the joint law under the universal version K^g , the second uses eq. (22), and the third is the change-of-variables formula for the pushforward measure $\mu_s = f_{\#}Q_s$. Hence by the uniqueness part of lemma 1, for every $s \in \mathcal{S}$ κ is a version of r_s . $\square$

Proof of Lemma 3.

Lemma 3 (a) The kernel of eq. (24) does not depend on s , and by independence of the detector noise ϵ it is a version of the conditional law of Y given T under both generators.

Lemma 3 (b) Under generator s , Z has density $(1 - \pi_s)\phi(z) + \pi_s\phi(z - a)$, which is strictly positive everywhere on $\mathbb{R}$.

Lemma 3 (c) Under generator s , the joint density of (Z, U) with respect to the product of Lebesgue and counting measure on $\mathbb{R} \times \{0, 1\}$ is $(1 - \pi_s)\phi(z)$ at $u = 0$ and $\pi_s\phi(z - a)$ at $u = 1$, so $\mathbb{P}_s(U = 1 \mid Z = z) = w_s(z)$. Given $Z = z$ and $U = u$, the kinematic variable equals $z - au$, so with $g = \text{id}$, $X = Y \sim \mathcal{N}(z - au, \sigma^2)$. Mixing over u yields the stated Gaussian mixture for $r_s(\cdot \mid z)$.

For fixed $z \in \mathbb{R}$, let $c = \phi(z - a) > 0$ and $e = \phi(z) > 0$. Consider the map

$$\xi : (0, 1) \rightarrow (0, 1), \quad \pi \mapsto \frac{\pi c}{\pi c + (1 - \pi)e}. \quad (68)$$

Its derivative is

$$\xi'(\pi) = \frac{ce}{[\pi c + (1 - \pi)e]^2} > 0, \quad (69)$$

so ξ is strictly increasing on $(0, 1)$. Since $\pi_H \neq \pi_P$ and $w_s(z) = \xi(\pi_s)$, it follows that $w_H(z) \neq w_P(z)$. The expectation of $r_s(\cdot \mid z)$ is $z - a w_s(z)$. Since $a \neq 0$, the means differ:

$$z - a w_H(z) \neq z - a w_P(z), \quad (70)$$

and hence the measures $r_H(\cdot \mid z)$ and $r_P(\cdot \mid z)$ differ for every $z \in \mathbb{R}$.

Lemma 3 (d) Under either generator, $Y = V + \epsilon$ with $V \sim \mathcal{N}(0, 1)$ and $\epsilon \sim \mathcal{N}(0, \sigma^2)$ independent, so $KQ_s = \mathcal{N}(0, 1 + \sigma^2)$. The mean of $f_{\#}Q_s$ is $a\pi_s$, and since $\pi_H \neq \pi_P$ and $a \neq 0$, $f_{\#}Q_H \neq f_{\#}Q_P$.

□

Proof of Proposition 2.

Proposition 2 (a) In example 1, the full response is universal by lemma 3 (a). By lemmata 3 (b) and 3 (c), the induced responses differ at every point of the common support $\mathcal{Z}_{\cap} = \mathbb{R} \implies \nu\{z \in \mathcal{Z}_{\cap} \mid r_H(\cdot \mid z) = r_P(\cdot \mid z)\} = 0$. Hence, by remark 1, the induced response is not universal.

Proposition 2 (b) Let $\mathcal{T} = \mathcal{Y} = \mathbb{R}$, $Q_H = Q_P = \mathcal{N}(0, 1)$, $K_H(\cdot \mid t) = \mathcal{N}(t, 1)$, and $K_P(\cdot \mid t) = \mathcal{N}(t, 4)$. These kernels differ at every $t \in \mathbb{R}$, and the support of Q_s is all of $\mathbb{R}$, so the full response is not universal.

Now let $f = \text{id}_{\mathbb{R}}$ and let $g(y) \equiv x_0$ for some fixed $x_0 \in \mathcal{X}$. Then $X = g(Y) = x_0$ almost surely under both generators, so $r_s(\cdot \mid z) = \delta_{x_0}$ for every $z \in \mathcal{Z}$ and both $s \in \mathcal{S}$. The induced responses are therefore identical.

□

Proof of Corollary 1. Apply lemma 2 with $\tilde{T} = (T, J)$ in the role of the full particle-level state, projection $f : (t, j) \mapsto t$, and $g = \text{id}_{\mathcal{Y}}$ so that $\tilde{K}^g = \tilde{K}$. The omitted variable of eq. (17) is $U = J$, and eq. (17) reduces directly to eq. (29).

Necessity If the conditional laws agree $\overline{Q}$ -a.e. on $\mathcal{T}_{\cap}$, then by eq. (29) for $\overline{Q}$ -almost every $t \in \mathcal{T}_{\cap}$, their mixtures satisfy $K_H(\cdot \mid t) = K_P(\cdot \mid t)$ which is the definition of universality across $\mathcal{S}$; by contraposition, disagreement on a set of positive $\overline{Q}$ -measure is necessary for non-universality.

Sufficiency under fibre injectivity The fibre of f over t is $\{t\} \times \mathcal{J} \cong \mathcal{J}$. When the fibre operator $\eta \mapsto \int_{\mathcal{J}} \tilde{K}(\cdot \mid t, j) \eta(dj)$ is injective for $\overline{Q}$ -almost every $t \in \mathcal{T}_{\cap}$, the mapping from conditional laws to induced responses is injective, so theorem 1 (c) guarantees that $K_H(\cdot \mid t) \neq K_P(\cdot \mid t)$ whenever $p_H(\cdot \mid t) \neq p_P(\cdot \mid t)$, establishing non-universality.

□

Proof of Proposition 3.

Proposition 3 (a) $\implies$ proposition 3 (b) Let r be a common version. Under the balanced mixture,

$$\begin{aligned} \forall A \in \mathcal{B}(\mathcal{X}) \quad \forall B \in \mathcal{B}(\mathcal{Z}) \quad \forall s \in \mathcal{S} \\ \mathbb{P}(S = s, Z \in B, X \in A) = \frac{1}{2} \int_B r(A \mid z) \mu_s(dz) = \mathbb{E}[\mathbb{1}_{S=s} \mathbb{1}_{Z \in B} r(A \mid Z)]. \end{aligned} \quad (71)$$

Hence $r(A \mid Z)$ is a version of $\mathbb{P}(X \in A \mid Z, S)$. Since it is $\sigma(Z)$ -measurable, it is also a version of $\mathbb{P}(X \in A \mid Z)$, and $\mathbb{P}(X \in A \mid Z, S) = \mathbb{P}(X \in A \mid Z)$ almost surely for every $A \in \mathcal{B}(\mathcal{X})$. This is equivalent to $X \perp\!\!\!\perp S \mid Z$ [7].

Proposition 3 (b) $\implies$ proposition 3 (a) Let ρ be a regular version of the conditional law $\mathbb{P}(X \mid Z)$, which exists by assumption 1. Conditional independence gives $\forall A \in \mathcal{B}(\mathcal{X}) \quad \mathbb{P}(X \in A \mid Z, S) = \rho(A \mid Z)$ almost surely. Hence,

$$\begin{aligned} \forall A \in \mathcal{B}(\mathcal{X}) \quad \forall B \in \mathcal{B}(\mathcal{Z}) \quad \forall s \in \mathcal{S} \\ \frac{1}{2} \mathbb{P}_s(Z \in B, X \in A) &= \mathbb{P}(S = s, Z \in B, X \in A) \\ &= \mathbb{E}[\mathbb{1}_{S=s} \mathbb{1}_{Z \in B} \rho(A \mid Z)] \\ &= \mathbb{P}(S = s) \mathbb{E}_s[\mathbb{1}_B(Z) \rho(A \mid Z)] = \frac{1}{2} \int_B \rho(A \mid z) \mu_s(dz). \end{aligned} \tag{72}$$

Multiplying both sides by 2 yields $\mathbb{P}_s(Z \in B, X \in A) = \int_B \rho(A \mid z) \mu_s(dz)$, so ρ is a version of both r_H and r_P .

Proposition 3 (b) $\iff$ proposition 3 (c) Since $d_{\text{KL}} \geq 0$ with equality almost surely if and only if its arguments coincide, $I(S; X \mid Z) = \mathbb{E}[d_{\text{KL}}(\eta_{(Z,X)} \parallel \eta_Z)] = 0$ if and only if $\eta_{(Z,X)} = \eta_Z$ almost surely. Because S is binary, this identity is equivalent to $\mathbb{P}(S \in \cdot \mid Z, X) = \mathbb{P}(S \in \cdot \mid Z)$ almost surely, which states $S \perp\!\!\!\perp X \mid Z$ [7], or equivalently $X \perp\!\!\!\perp S \mid Z$ by symmetry.

□

Proof of Proposition 4. Under the balanced mixture $\mathbb{P}$, $\mathbb{P}(S = H, Z \in B) = \frac{1}{2} \mu_H(B) = \int_B \frac{1}{2} m_H \, d\nu$, and the marginal law of Z is ν ; hence $\eta(Z) = \frac{1}{2} m_H(Z)$ is a version of η_Z , and $1 - \eta(Z) = \frac{1}{2} m_P(Z)$ ν -almost surely. Let $\eta_H(z) = \eta(z)$ and $\eta_P(z) = 1 - \eta(z)$.

$\forall z \in \mathcal{Z} \quad \eta_s(z) > 0 \implies \bar{r}(\cdot \mid z)$ is a version of the conditional law of $X \mid Z = z$, and $r_s(\cdot \mid z) \ll \bar{r}(\cdot \mid z)$. By Doob's theorem on measurable densities [7],

$$\begin{aligned} \forall s \in \mathcal{S} \quad \exists \rho_s: \mathcal{Z} \times \mathcal{X} \rightarrow [0, \infty) \quad \eta_s(z) r_s(dx \mid z) &= \eta_s(z) \rho_s(z, x) \bar{r}(dx \mid z) \\ \wedge \eta_H(z) \rho_H(z, x) + \eta_P(z) \rho_P(z, x) &= 1 \quad \bar{r}(\cdot \mid z)\text{-a.e.} \end{aligned} \tag{73}$$

$$\forall A \in \mathcal{B}(\mathcal{X}) \quad \forall B \in \mathcal{B}(\mathcal{Z}) \quad \mathbb{P}(S = H, Z \in B, X \in A) = \int_B \eta_H(z) \int_A \rho_H(z, x) \bar{r}(dx \mid z) \nu(dz), \tag{74}$$

so that $\eta_{(Z,X)} = \eta_H(Z) \rho_H(Z, X)$ and $1 - \eta_{(Z,X)} = \eta_P(Z) \rho_P(Z, X)$ almost surely. Consequently, on $\mathcal{Z}_\cap$ where $\eta_s(Z) > 0$, the likelihood ratios satisfy

$$\frac{\eta_{(Z,X)}}{\eta_Z} = \rho_H(Z, X) \quad \text{and} \quad \frac{1 - \eta_{(Z,X)}}{1 - \eta_Z} = \rho_P(Z, X). \tag{75}$$

Substituting into eq. (35) and conditioning on $Z = z \in \mathcal{Z}_\cap$,

$$\begin{aligned}
\mathbb{E}[d_{\text{KL}}(\eta_{(Z,X)} \parallel \eta_Z) \mid Z = z] &= \sum_{s \in \mathcal{S}} \eta_s(z) \int_{\mathcal{X}} \rho_s(z, x) \ln \left(\frac{\eta_s(z) \rho_s(z, x)}{\eta_s(z)} \right) \bar{r}(\mathrm{d}x \mid z) \\
&= \sum_{s \in \mathcal{S}} \eta_s(z) \int_{\mathcal{X}} \rho_s(z, x) \ln \rho_s(z, x) \bar{r}(\mathrm{d}x \mid z) \\
&= \sum_{s \in \mathcal{S}} \eta_s(z) D_{\text{KL}}(r_s(\cdot \mid z) \parallel \bar{r}(\cdot \mid z)) \\
&= \text{JS}_{\eta(z)}(r_H(\cdot \mid z), r_P(\cdot \mid z)).
\end{aligned} \tag{76}$$

Off $\mathcal{Z}_\cap$, one of $\eta_H(z)$, $\eta_P(z)$ vanishes and $\bar{r}(\cdot \mid z)$ coincides with the other induced response, so the integrand vanishes identically. Integrating over $\mathcal{Z}$ against ν reduces to the integral over $\mathcal{Z}_\cap$ and yields eq. (38). $\square$

Proof of Proposition 5. Conditional on W , the label S is Bernoulli with parameter η_W (with $S = H$ coded as 1), so that

$$\begin{aligned}
\mathbb{E}[\ell\{S, D(W)\} \mid W] &= -\eta_W \ln D(W) - (1 - \eta_W) \ln[1 - D(W)] \\
&= h(\eta_W) + d_{\text{KL}}[\eta_W \parallel D(W)],
\end{aligned} \tag{77}$$

where $d_{\text{KL}}(p \parallel q) = p \ln \frac{p}{q} + (1 - p) \ln \frac{1-p}{1-q}$ is extended continuously to the boundaries of $[0, 1]$.

Both terms on the right-hand side are non-negative, so taking unconditional expectations is justified and yields eq. (41). The inequality and its equality condition follow immediately because $h(\eta_W) \geq 0$ and $d_{\text{KL}}(p \parallel q) \geq 0$ with $d_{\text{KL}}(p \parallel q) = 0 \iff p = q$.

For the identity, let $\eta_{ZX} = \eta_{(Z,X)}$ and expand the definition of d_{KL} :

$$d_{\text{KL}}(\eta_{ZX} \parallel \eta_Z) = -h(\eta_{ZX}) - \eta_{ZX} \ln \eta_Z - (1 - \eta_{ZX}) \ln(1 - \eta_Z). \tag{78}$$

Since $-\ln \eta_Z$ and $-\ln(1 - \eta_Z)$ are non-negative and $\sigma(Z)$ -measurable, and $\mathbb{E}[\eta_{ZX} \mid Z] = \eta_Z$ almost surely by the tower property of conditional expectation,

$$\begin{aligned}
\mathbb{E}[-\eta_{ZX} \ln \eta_Z - (1 - \eta_{ZX}) \ln(1 - \eta_Z)] &= \mathbb{E}\left[\mathbb{E}\left\{-\eta_{ZX} \ln \eta_Z - (1 - \eta_{ZX}) \ln(1 - \eta_Z) \mid Z\right\}\right] \\
&= \mathbb{E}[-\eta_Z \ln \eta_Z - (1 - \eta_Z) \ln(1 - \eta_Z)] \\
&= H(S \mid Z) \\
&\leq \ln 2.
\end{aligned} \tag{79}$$

Taking expectations on both sides of the expansion yields

$$\mathbb{E}[d_{\text{KL}}(\eta_{ZX} \parallel \eta_Z)] = H(S \mid Z) - \mathbb{E}[h(\eta_{ZX})] = H(S \mid Z) - H(S \mid Z, X) = I(S; X \mid Z), \tag{80}$$

where all terms are finite. Finally, eq. (42) follows from eq. (41) evaluated at the Bayes-optimal choices $D_Z^* = \eta_Z$ and $D_{ZX}^* = \eta_{ZX}$, for which the d_{KL} terms vanish identically. $\square$

Proof of Corollary 2. By eq. (41), $L_Z(D_Z) = H(S | Z) + \varepsilon_Z$ and $L_{ZX}(D_{ZX}) = H(S | Z, X) + \varepsilon_{ZX}$, where $\varepsilon_Z = \mathbb{E}[d_{\text{KL}}\{\eta_Z \| D_Z(Z)\}] \geq 0$ and $\varepsilon_{ZX} = \mathbb{E}[d_{\text{KL}}\{\eta_{ZX} \| D_{ZX}(Z, X)\}] \geq 0$. Subtracting these two identities and substituting $H(S | Z) - H(S | Z, X) = I(S; X | Z)$ from proposition 5 yields eq. (43). Corollaries 2 (a) to 2 (c) then follow immediately from eq. (43) together with the non-negativity of ε_Z , ε_{ZX} , and $I(S; X | Z)$. $\square$

Proof of Proposition 6. Since the test sample is stratified with $n_H = n_P = n/2$, the estimator decomposes as

$$\hat{\delta} = \frac{1}{2}(\bar{d}_H + \bar{d}_P), \quad \bar{d}_s = \frac{1}{n_s} \sum_{i: S_i=s} d_i. \quad (81)$$

Conditional on $\hat{D}_Z$ and $\hat{D}_{ZX}$, the events within stratum s are independent and identically distributed with conditional mean $\mathbb{E}[d_i | S_i = s]$ and finite variance σ_s^2 , and the two strata are independent. Therefore,

$$\mathbb{E}[\hat{\delta} | \hat{D}_Z, \hat{D}_{ZX}] = \frac{1}{2}\mathbb{E}[d | S = H] + \frac{1}{2}\mathbb{E}[d | S = P] = \mathbb{E}[d] = \Delta(\hat{D}_Z, \hat{D}_{ZX}), \quad (82)$$

which is the population expectation of d under the balanced mixture $\mathbb{P}$. For the conditional variance, the independence of the two strata gives

$$\text{Var}(\hat{\delta} | \hat{D}_Z, \hat{D}_{ZX}) = \frac{1}{4} \text{Var}(\bar{d}_H) + \frac{1}{4} \text{Var}(\bar{d}_P) = \frac{1}{4} \left(\frac{\sigma_H^2}{n/2} \right) + \frac{1}{4} \left(\frac{\sigma_P^2}{n/2} \right) = \frac{\sigma_H^2 + \sigma_P^2}{2n}. \quad (83)$$

$\square$

Proof of Proposition 7.

Proposition 7 (a) By eq. (49), the true detector-level law is $\mathbb{P}_{\text{data}} = r_H \mu_H \neq r_P \mu_H$ by hypothesis.

$$\mu_H \ll \mu_P \implies \mu_H \in \mathcal{P}_P \implies \mu_H \notin \Theta_{\text{cl}}.$$

Proposition 7 (b)

$$\forall \mu \in \mathcal{P}_P \quad D_{\text{KL}}(\mathbb{P}_{\text{data}} \| r_P \mu) \geq 0 \wedge D_{\text{KL}}(\mathbb{P}_{\text{data}} \| r_P \mu) = 0 \iff r_P \mu = \mathbb{P}_{\text{data}}. \quad (84)$$

Therefore,

$$\mu \in \Theta_{\text{cl}} \iff r_P \mu = \mathbb{P}_{\text{data}} \iff D_{\text{KL}}(\mathbb{P}_{\text{data}} \| r_P \mu) = 0. \quad (85)$$

$$\Theta_{\text{cl}} \neq \emptyset \implies \min_{\mu \in \mathcal{P}_P} D_{\text{KL}}(\mathbb{P}_{\text{data}} \| r_P \mu) = 0 \wedge \underset{\mu \in \mathcal{P}_P}{\text{argmin}} D_{\text{KL}}(\mathbb{P}_{\text{data}} \| r_P \mu) = \Theta_{\text{cl}}. \quad (86)$$

Since $\mu_H \notin \Theta_{\text{cl}}$ by proposition 7 (a), no population maximum-likelihood solution can coincide with the true particle-level law μ_H . $\square$

Proof of Theorem 2.

Theorem 2 (a) $\forall \Delta_0 = d(f_{\#}Q_1, f_{\#}Q_2) > 0 \quad KQ_1 = KQ_2$

$\implies$ the observable sample $Y_{1:n} = (Y_1, \dots, Y_n)$ has the identical joint law $\mathbb{P}_{Y_{1:n}} = (KQ_1)^{\otimes n} = (KQ_2)^{\otimes n}$ under both Q_1 and Q_2

$\implies$ the estimator $\hat{\phi}_n(Y_{1:n})$ has the identical sampling distribution under both hypotheses.
 $\forall i \in \{1, 2\} \quad \forall E_i = \left\{ d(\hat{\phi}_n, f_{\#}Q_i) < \Delta_0/2 \right\}$, by the triangle inequality,

$$d(f_{\#}Q_1, f_{\#}Q_2) \leq d(\hat{\phi}_n, f_{\#}Q_1) + d(\hat{\phi}_n, f_{\#}Q_2) \implies E_1 \cap E_2 = \emptyset. \quad (87)$$

Therefore, under the common sampling law,

$$\mathbb{P}_{Q_1}(E_1) + \mathbb{P}_{Q_2}(E_2) = \mathbb{P}_{Q_1}(E_1) + \mathbb{P}_{Q_1}(E_2) = \mathbb{P}_{Q_1}(E_1 \cup E_2) \leq 1. \quad (88)$$

$$\implies \max_{i \in \{1, 2\}} \mathbb{P}_{Q_i}(E_i^c) \geq \frac{1}{2}$$

$$\implies \hat{\phi}_n \text{ cannot converge in probability to both } f_{\#}Q_1 \text{ under } Q_1 \text{ and } f_{\#}Q_2 \text{ under } Q_2.$$

Theorem 2 (b)

$$\forall Q \in \mathcal{Q} \quad D_{\text{KL}}(KQ_1 \parallel KQ) \geq 0 \quad \wedge \quad D_{\text{KL}}(KQ_1 \parallel KQ) = 0 \iff KQ = KQ_1. \quad (89)$$

$$\therefore \underset{Q \in \mathcal{Q}}{\operatorname{argmin}} D_{\text{KL}}(KQ_1 \parallel KQ) = \{Q \in \mathcal{Q} \mid KQ = KQ_1\}. \quad (90)$$

By eq. (53), $KQ_2 = KQ_1 \implies Q_2$ achieves the global minimum value of 0. Since $f_{\#}Q_2 \neq f_{\#}Q_1$, population likelihood maximization does not isolate $f_{\#}Q_1$.

□

Proof of Proposition 8. By the triangle inequality,

$$\forall \Delta_0 = d(f_{\#}Q_1, f_{\#}Q_2) \quad \forall P_i = (KQ_i)^{\otimes n} \quad \forall E = \left\{ d(\hat{\phi}_n, f_{\#}Q_1) < \frac{\Delta_0}{2} \right\} \quad (91)$$

$$d(\hat{\phi}_n, f_{\#}Q_2) > \frac{\Delta_0}{2}.$$

$$\therefore E \subseteq \left\{ d(\hat{\phi}_n, f_{\#}Q_2) > \frac{\Delta_0}{2} \right\} \subseteq \left\{ d(\hat{\phi}_n, f_{\#}Q_2) \geq \frac{\Delta_0}{2} \right\}. \quad (92)$$

Therefore,

$$P_1 \left[d(\hat{\phi}_n, f_{\#}Q_1) \geq \frac{\Delta_0}{2} \right] + P_2 \left[d(\hat{\phi}_n, f_{\#}Q_2) \geq \frac{\Delta_0}{2} \right] \geq P_1(E^c) + P_2(E) \geq 1 - d_{\text{TV}}(P_1, P_2). \quad (93)$$

The maximum of the two probabilities on the left is at least half of their sum:

$$\max_{i \in \{1,2\}} P_i \left[d(\hat{\phi}_n, f_{\#} Q_i) \geq \frac{\Delta_0}{2} \right] \geq \frac{1}{2} - \frac{1}{2} d_{\text{TV}}(P_1, P_2). \quad (94)$$

Finally, $d_{\text{TV}}(P_1, P_2) \leq n d_{\text{TV}}(KQ_1, KQ_2)$ by subadditivity of total variation under product measures [22], which yields eq. (56). $\square$

Proof of Proposition 9. Consider the latent-species model of example 1, where $T = (V, U) \in \mathbb{R} \times \{0, 1\}$, U represents the unobserved species indicator, $Z = f(V, U) = V + aU$ is the particle-level mass summary with $a \neq 0$, and $g = \text{id}_{\mathbb{R}}$ so that $X = Y = V + \epsilon$. Proposition 9 (a) is established by lemma 3 (a). Proposition 9 (b) follows from lemmata 3 (b) and 3 (c), where $\mathcal{Z}_{\cap} = \mathbb{R}$ and the induced kernels differ for all $z \in \mathbb{R}$. Proposition 9 (c) is established by lemma 3 (d) and theorem 2, choosing any model class $\mathcal{Q} \supseteq \{Q_H, Q_P\}$. All three properties are simultaneously driven by the dependence of Z on U , the conditional independence of the detector observable Y from U given V (erasure of species by K), and the prior fraction discrepancy $\pi_H \neq \pi_P$. $\square$

Proof of Proposition 10. Let $N \in \mathbb{N}$ denote the sample size drawn from $\mathbb{P}_P$, and let $\zeta : (\mathcal{T} \times \mathcal{Y})^N \times \Omega_{\text{aux}} \rightarrow [0, 1]$ be any test function based on Φ and auxiliary randomness $\omega \sim \mathbb{P}_{\text{aux}}$.

Because Φ is a measurable function of a sample with law $\mathbb{P}_P^{\otimes N} \otimes \mathbb{P}_{\text{aux}}$, its sampling distribution depends exclusively on $\mathbb{P}_P$ and $\mathbb{P}_{\text{aux}}$, and is invariant to $\mathbb{P}_H$. Therefore, under any two specifications $(\mathbb{P}_H, \mathbb{P}_P)$ and $(\mathbb{P}'_H, \mathbb{P}_P)$ that share $\mathbb{P}_P$,

$$\mathbb{E}_{(\mathbb{P}_H, \mathbb{P}_P)}[\zeta] = \mathbb{E}_{(\mathbb{P}'_H, \mathbb{P}_P)}[\zeta]. \quad (95)$$

Choosing $(\mathbb{P}_H, \mathbb{P}_P)$ to satisfy H_0 with size $\alpha := \mathbb{E}_{(\mathbb{P}_H, \mathbb{P}_P)}[\zeta]$, the rejection probability under any alternative $(\mathbb{P}'_H, \mathbb{P}_P)$ is identically α . The power against any such alternative is therefore equal to the size. $\square$

Proof of Corollary 3. Let $(\mathcal{C}, \mathcal{B}(\mathcal{C}))$ be the standard Borel space in which $C = c(T)$ takes values. By hypothesis

$$\exists \kappa : \mathcal{Z} \times \mathcal{C} \rightarrow \mathcal{P}(\mathcal{X}) \quad \forall s \in \mathcal{S} \quad Q_s \left\{ t \in \mathcal{T} \mid K^g(\cdot \mid t) = \kappa(\cdot \mid f(t), c(t)) \right\} = 1. \quad (96)$$

Applying proposition 1 with the augmented summary $(f, c) : \mathcal{T} \rightarrow \mathcal{Z} \times \mathcal{C}$ in place of f , the response induced on $((f, c), g)$ is universal across $\mathcal{S}$ with common version κ . Applying proposition 3 with the conditioning variable (Z, C) in place of Z , universality of the induced response is equivalent to $X \perp\!\!\!\perp S \mid (Z, C)$, which by proposition 3 (c) holds iff $I(S; X \mid Z, C) = 0$. $\square$

Proof of Corollary 4.

Corollary 4 (a) Set $f = \text{id}_{\mathcal{T}}$ and $g = \text{id}_{\mathcal{Y}}$, so that $Z = T$ and $X = Y$. The induced response on (f, g) coincides identically with the full response K . The equivalence between $I(S; Y \mid T) = 0$ and universality of K across $\mathcal{S}$ follows directly from proposition 3 (c).

Corollary 4 (b) Set $f = \text{id}_{\mathcal{T}}$ and retain the detector-level projection $g: \mathcal{Y} \rightarrow \mathcal{X}$, so that $Z = T$ and $X = g(Y)$. The induced response on (f, g) is the g -projected full response K^g , and the equivalence follows likewise from proposition 3 (c).

For the inequality, because $X = g(Y)$ is $\sigma(Y)$ -measurable, $\sigma(X, Y) = \sigma(Y)$. The chain rule for conditional mutual information [18] yields

$$I(S; Y \mid T) = I(S; (X, Y) \mid T) = I(S; X \mid T) + I(S; Y \mid T, X). \quad (97)$$

Since conditional mutual information is non-negative, $I(S; Y \mid T, X) \geq 0$, which establishes $I(S; X \mid T) \leq I(S; Y \mid T)$.

□